

Q1 2021 (31 Aug Shift 2)

The mean and variance of 7 observations are 8 and 16 respectively. If two observations are 6 and 8, then the variance of the remaining 5 observations is :

(1) $\frac{92}{5}$

(2) $\frac{134}{5}$

(3) $\frac{536}{25}$

(4) $\frac{112}{5}$

Q2 2021 (31 Aug Shift 1)

The mean of 10 numbers

$7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$ is .

Q3 2021 (27 Aug Shift 2)

An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to .

Q4 2021 (27 Aug Shift 1)

Let n be an odd natural number such that the variance of $1, 2, 3, 4, \dots, n$ is 14. Then n is equal to .

Q5 2021 (26 Aug Shift 2)

Let the mean and variance of four numbers 3, 7, x and y ($x > y$) be 5 and 10 respectively. Then the mean of four numbers $3 + 2x, 7 + 2y, x + y$ and $x - y$ is.

Q6 2021 (26 Aug Shift 1)

The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is:

(1) (11, 26)

(2) (10.5, 25)

(3) (11, 25)

(4) (10.5, 26)

Q6 (4)

Q1 (3)

Let 8, 16, x_1, x_2, x_3, x_4, x_5 be the observations.

$$\text{Now } \frac{x_1 + x_2 + \dots + x_5 + 14}{7} = 8$$

$$\Rightarrow \sum_{i=1}^5 x_i = 42 \dots (1)$$

$$\text{Also } \frac{x_1^2 + x_2^2 + \dots + x_5^2 + 8^2 + 6^2}{7} - 64 = 16$$

$$\Rightarrow \sum_{i=1}^5 x_i^2 = 560 - 100 = 460 \dots (2)$$

So variance of $x_1, x_2, \dots, x_5$

$$= \frac{460}{5} - \left(\frac{42}{5}\right)^2 = \frac{2300 - 1764}{25} = \frac{536}{25}$$

Q2 (398)

$$7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14 \dots \dots$$

$$T_n = (3n + 4)(2n + 6) = 2(3n + 4)(n + 3)$$

$$= 2(3n^2 + 13n + 12) = 6n^2 + 26n + 24$$

$$S_{10} = \sum_{n=1}^{10} T_n = 6 \sum_{n=1}^{10} n^2 + 26 \sum_{n=1}^{10} n + 24 \sum_{n=1}^{10} 1$$

$$= \frac{6(10 \times 11 \times 21)}{6} + 26 \times \frac{10 \times 11}{2} + 24 \times 10$$

$$= 10 \times 11(21 + 13) + 240$$

$$= 3980$$

$$\text{Mean} = \frac{S_{10}}{10} = \frac{3980}{10} = 398.$$

Q3 (25)

$$\sigma_b^2 = 2 \text{ (variance of boys) } n_1 = \text{no. of boys}$$

$$\bar{x}_b = 12 \quad n_2 = \text{no. of girls}$$

$$\sigma_g^2 = 2$$

$$\bar{x}_g = \frac{50 \times 15 - 12 \times \sigma_b}{30} = \frac{750 - 12 \times 20}{30} = 17 = \mu$$

variance of combined series

$$\sigma^2 = \frac{n_1 \sigma_b^2 + n_2 \sigma_g^2}{n_1 + n_2} + \frac{n_1 \cdot n_2}{(n_1 + n_2)^2} (\bar{x}_b - \bar{x}_g)^2$$

$$\sigma^2 = \frac{20 \times 2 + 30 \times 2}{20 + 30} + \frac{20 \times 30}{(20 + 30)^2} (12 - 17)^2$$

$$\sigma^2 = 8.$$

$$\Rightarrow \mu + \sigma^2 = 17 + 8 = 25$$

Q4 (13)

$$\frac{n^2-1}{12} = 14 \Rightarrow n = 13$$

Q5 (12)

$$5 = \frac{3+7+x+y}{4} \Rightarrow x + y = 10$$

$$\text{Var}(x) = 10 = \frac{3^2+7^2+x^2+y^2}{4} - 25$$

$$140 = 49 + 9 + x^2 + y^2$$

$$x^2 + y^2 = 82$$

$$x + y = 10$$

$$\Rightarrow (x, y) = (9, 1)$$

Four numbers are 21, 9, 10, 8

$$\text{Mean} = \frac{48}{4} = 12$$

Q6 (4)

Given:

$$\text{Mean}(\bar{x}) = \frac{\sum x_i}{20} = 10$$

$$\text{or } \sum x_i = 200 \text{ (incorrect)}$$

$$\text{or } 200 - 25 + 35 = 210 = \sum x_i \text{ (Correct)}$$

$$\text{Now correct } \bar{x} = \frac{210}{20} = 10.5$$

again given S.D = 2.5(σ)

$$\sigma^2 = \frac{\sum x_i^2}{20} - (10)^2 = (2.5)^2$$

$$\text{or } \sum x_i^2 = 2125 \text{ (incorrect)}$$

$$\text{or } \sum x_i^2 = 2125 - 25^2 + 35^2$$

$$= 2725 \text{ (Correct)}$$

$$\therefore \text{correct } \sigma^2 = \frac{2725}{20} - (10.5)^2$$

$$\sigma^2 = 26$$

$$\text{or } \sigma = 26$$

$$\therefore \alpha = 10.5, \beta = 26$$